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Supply Chain Network Optimization & Simulation Model Digital Twin

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Executive Summary

About Borouge: Borouge is a leading petrochemical company that provides creative plastics solutions for a wide range of industries. Established in 1998, Borouge is a joint venture between the Abu Dhabi National Oil Company (ADNOC), one of the world's leading oil and gas companies and Austria based Borealis. petrochemical and plastics manufacturing plant is in Al Ruwais at about 250 km west of Abu Dhabi. In addition to our production site in Abu Dhabi, Borouge also has a Compounding Manufacturing Plant (CMP) in Shanghai which compounds resins from the Ruwais plant specifically for the Asian automotive industry.

Supply Chain Design & Planning Model:
Need
As a world leader in polymer solutions, we required an established digital twin for its supply chain planning (more specifically logistic network optimization & simulation) and configurable assets. The initial transformation process focused on UAE production site, global supply chain network and operations, with an emphasis on cost and service levers. The need was to develop a digital twin within the first 8 months of the project. The solution was to be used as a strategic tool to define the optimal network. This was to be done by selecting the optimal Inventory Holding Point ("IHP") to serve customers based on associated costs, capacities and lead times.
Key Objectives
The primary objective of the Supply Chain Network Optimization & Simulation model was to optimize the Inventory Holding Point ("IHP") selection between Al Ruwais and the destination port to minimize network cost leading to better

a. Visibility
b. Capability
c. Autonomy
d. Cost Savings

Development Brief
The model was developed using:

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a. A baseline based on 12 months production plan, demand forecast, historic transport, network inventory position, planned site closures
b. An optimized network without following predefined flows in production plan
c. Transportation Modeling

    i. Packaging facility to UAE IHPs – road, sea freight and rail modeled
    ii. UAE IHPs to International IHPs – sea freight modeled
    iii. All IHPs to Destination Port – sea freight modeled
    iv. Destination Port to Customer – not modeled
d. Costs: Packing, Handling, Transport ($/ton), Storage ($/ton/mo)
e. Constraints: Packing capacity (tons/mo per packing line), transport lane capacity (tons/mo), inventory capacity (tons/mo at all IHPs) & transit times / service levels
f. Scenarios where existing constraints can be removed, modified and new constraints could be introduced
g. Frequency: Monthly cadence runs utilizing M0-M11 (12 months production plan and demand)
h. Processes included ingestion of data, cleansing and transformation of data, model building, scenarios building and transforming model output to user consumable reports.

Outcome

A Network Optimization Model utilizing applications like Supply Chain Guru X & Data Guru was created that lead to following outcomes and certain capabilities for the end user:

a. The model outputs enabled comparison of different scenarios based on criteria such as LVC (Logistics Variable Cost), DIT (Delay in Transit) utilization, & ending inventory
b. The model provided volumes, at a material level, to be shipped from Ruwais to IHPs and from IHPs to destination ports
c. The information was then subsequently shared to the relevant teams for execution
d. Adjustments to the model take the form of what is known as scenarios. Example scenarios/model recommendations included the following:

    i. Optimize IHP Selection - Allow all/specific customers to have IHP optimized
    ii. Modify SLA (Service Level Agreements) group ratios
    iii. Modify site throughput/storage capacity
    iv. Modify site packaging capacity/capability
    v. Close IHP/Open IHP
    vi. Alter transport costs
    vii. Modify port sourcing ratios by customer
    viii. Define volume ratio split between DIT and Warehouse IHPs at customer product level
    ix. Constraints applied at the calendar year level – E.x. minimum yearly site throughput
    x. Feeder leg ratio adjustments – The feeder leg is the lane between plant and UAE IHPs; used to adjust ratio of volume to be split between rail/sea
    xi. Ability to group destination ports and apply flow count constraints to limit the count of IHPs servicing all the ports in a group

Results

a. Model identified potential cost savings of up to 9% of supply chain logistics cost compared to the baseline for the 12-month period. The model outputs enable comparison of different scenarios based on criteria such as LVC, DIT utilization, ending inventory.

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b. The model provides volumes, at a material level, to be shipped from Al Ruwais to IHPs and from IHPs to destination ports.

This information has been utilized by various departments for orders executions. It is helping the business to better analyze data to optimize operations, enhance customer experience, and reduce costs.

Second-Level Heading.

Project Implementation

Supply Chain Network Overview. Global supply chain consists of a single plant and packaging facility at Al Ruwais and 22 Inventory Holding Points ("IHP") of which two have packaging facilities (Khalifa port and Nansha warehouse). These 22 IHPs then deliver to seaports for international customers and utilize a road network for local customers based in the middle east. From the 22 IHPs, 6 are locally situated within the UAE and are responsible for customers within the middle east. The ports are termed as Injection Points ("InPo") within the model and consist of 390 ports which then supply to 112 countries, including the UAE. An IHP can either be an RDC (regional distribution center), WH (warehouse), RDS (Regional Distribution Centre), DIT (Delay in Transit storage locations) or SOW (Sales on Water. A holistic view of the global supply chain network is shown below:

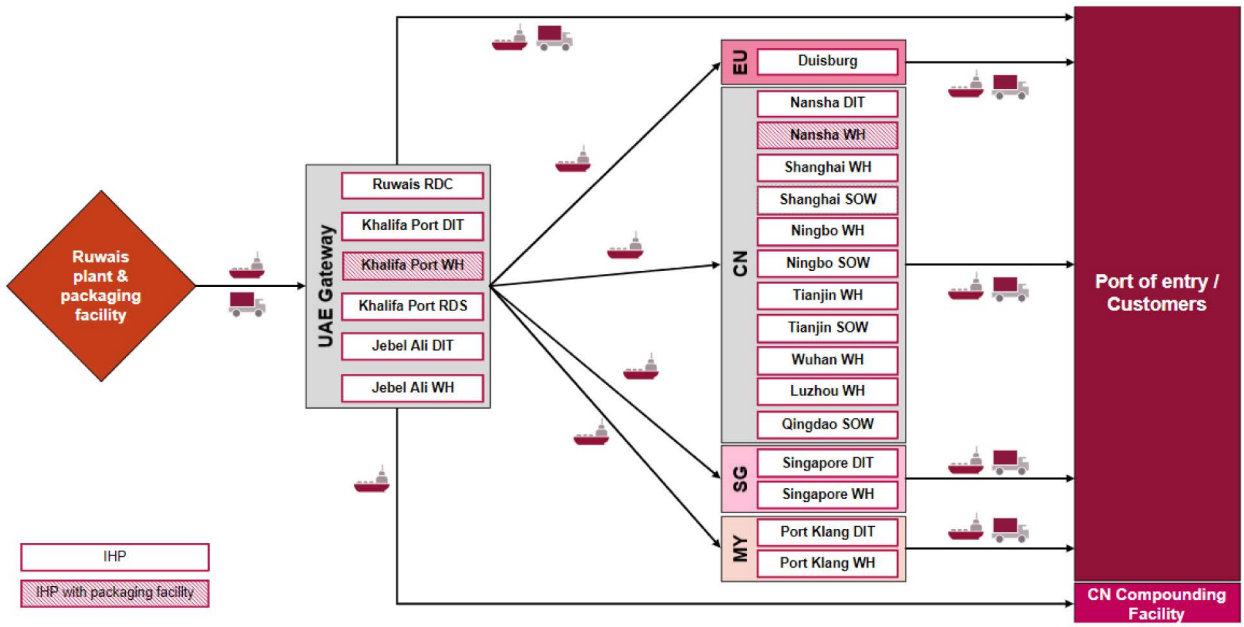

Figure 1—A view of the global supply network

Once the network was scoped, mapped out and understood, the next step was to implement a digital twin of global supply chain network using a supply chain network optimization software to arrive at an understanding of the total costs that the network incurs and harness the tool's optimization capabilities to further unlock cost savings opportunities. For this purpose, selected Applications Supply Chain Guru and Data Guru tools for implementing the digital twin of the network. A summary of these tools are given below.

Product Description. The supply chain network optimization & simulation model was created using Supply Chain and Data Guru suite of applications. A brief overview of the applications used is mentioned below.

Network Optimization and Simulation Tool. Supply Chain Guru ("SCGX") is a leading supply chain design and analysis application with an interactive interface, advanced supply chain intelligence database

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and design engine. System features include supply chain analysis algorithms and techniques, network optimization, multi-echelon inventory modeling, product flow path optimization, demand classification, discrete event simulation, transportation route and asset optimization, data editing, visualization and reporting. Finally, SCGX also has a cloud-based design platform (as opposed to the desktop version) that enables scalable scenario running and sharing across the enterprise. The global supply chain was modelled on the SCGX desktop software.

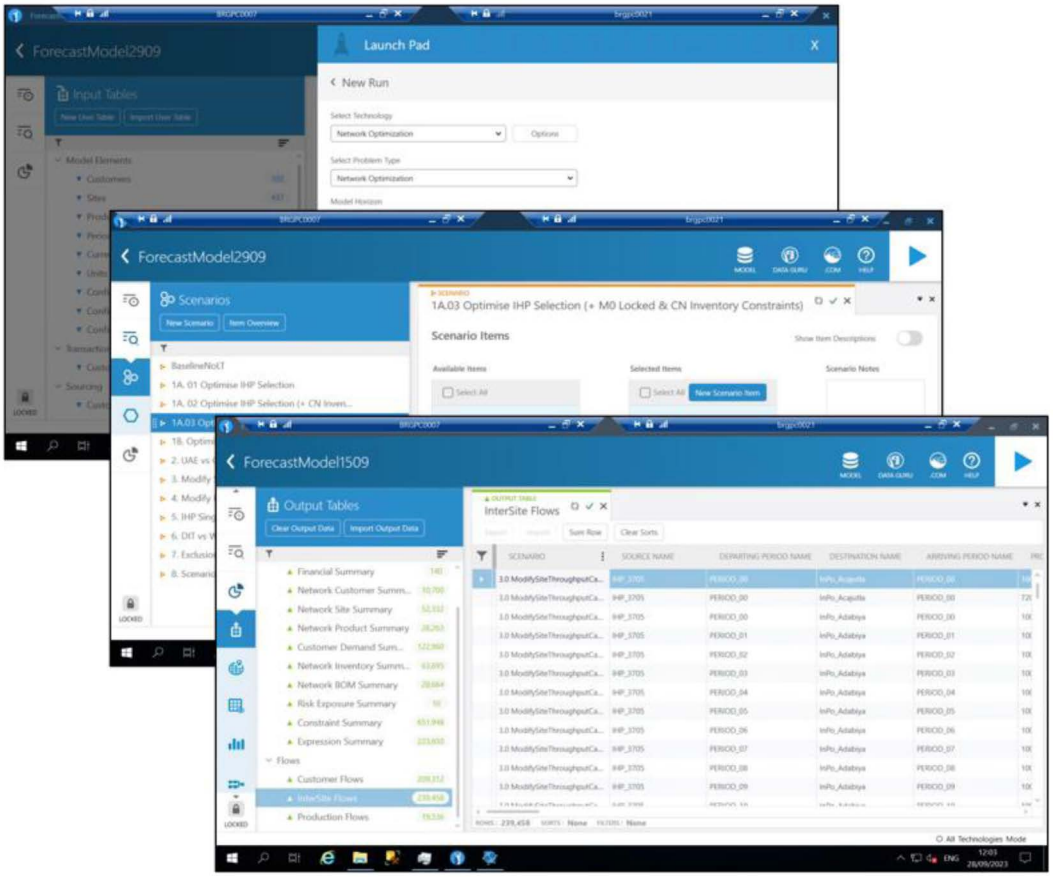

Figure 2—Screenshots of Supply Chain Guru interface

Data Pipeline & Transformation Tool. Data Guru ("DG") is a self-service drag-and-drop application empowering all users to connect and transform enterprise data from disparate systems. It simplifies data transformation and creates documented, repeatable, time-saving workflows. It is an interactive visual tool which removes coding and query-building from process and enables analysts to effectively work with big data. Its design toolkit has pre-built tools to guide the analyst through every step of the design process and it also provides the ability to debug models, analyze input and output, and apply advanced analytics.

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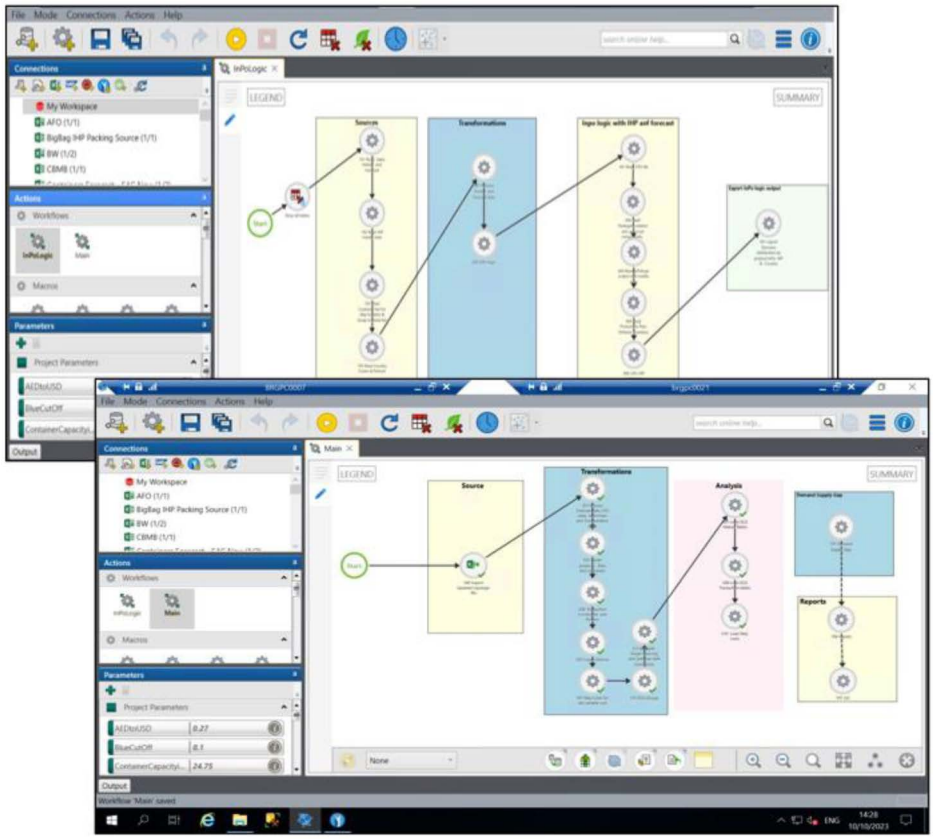

Figure 3—Screenshots of Data Guru interface

MODEL SETUP AND RESULTS

Variables and Constraints. A typical SCGX model consists of Customers, Sites, Products, Periods, Modes, Customer Demand, and other policy (Site Sourcing, Customer Sourcing, Production, etc.) and constraint tables (Flow, Inventory, Production, etc.). The Al Ruwais facility, 22 IHPs and 211 InPos were modeled as sites and keyed into the Sites table. Note that the original number of 390 InPos were reduced to 211 by a logic that selected the most used InPos for a given customer and product combination to reduce the complexity of the model, the process of which has been explained in the InPo Logic sub-section. The customer demand was aggregated at the country level, resulting in 112 different customers being entered into the Customers table. has a SKU list of around 400 to 500 SKUs, which were added to the Products table. The modes of transportation used for the global network consisted of ocean freight, rail and land-based trucking transports of various categories such as flatbed, sideboard, pick-up and box trucks. The transport costs were aggregated at origin/destination (some instances at origin/destination/transit time/carrier) to get an average $/ton cost.

The constraints in the model were based on limitations to the supply chain at either the nodes or the lanes between the nodes and were modeled accordingly using the constraints tables given in SCGX for that purpose. More on the various constraints and the tables used have been elaborated in the 'Creating the baseline model' sub-section.

MODEL ARCHITECTURE

The Optimization & Simulation tools (Data Guru - DG and Supply Chain Guru – SCGX) is deployed on a stand-alone Virtual Machine (VM) and the user is required to gather all the necessary input data and import it into the VM. The key benefit of SCGX is it allows the user to develop different scenarios for their network to identify the optimal setup for their upcoming year. The image below illustrates the model architecture:

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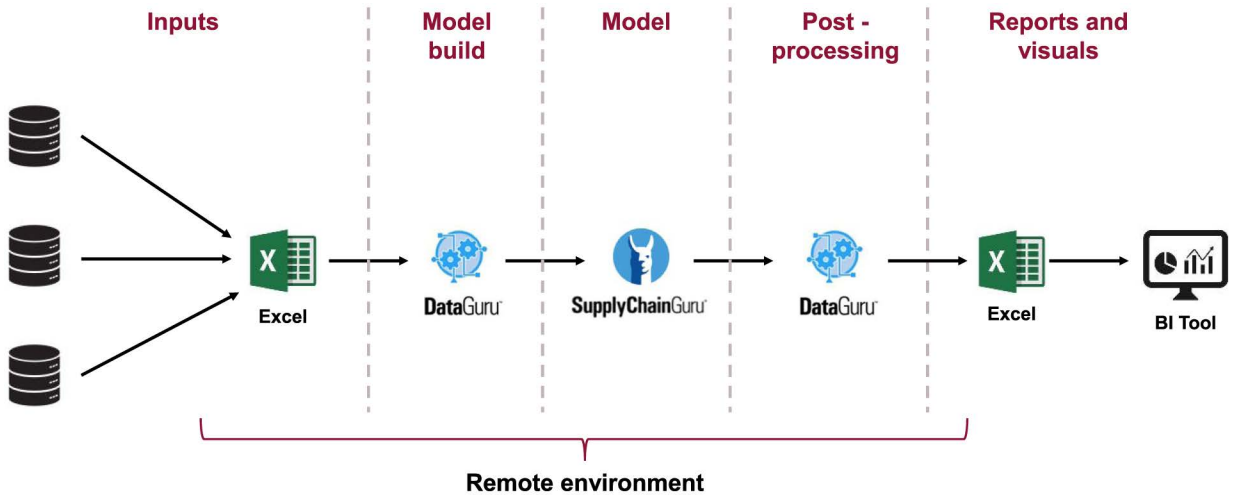

Figure 4—Model architecture of the network

As can be seen from the architecture, the source data files are transformed by DG into tables that are fit for consumption by SCGX. Once the model runs, the output tables are again transformed by a different SG script into excel files/reports that can be easily consumed by stakeholders. Finally, these reports can be converted to visualizations in a BI tool of choice as selected.

Data Ingestion & Transformation. The various sources of data are loaded into the VM by the user and then are combined and transformed into the input file formats required by the SCGX model. The solution workflow is shown below:

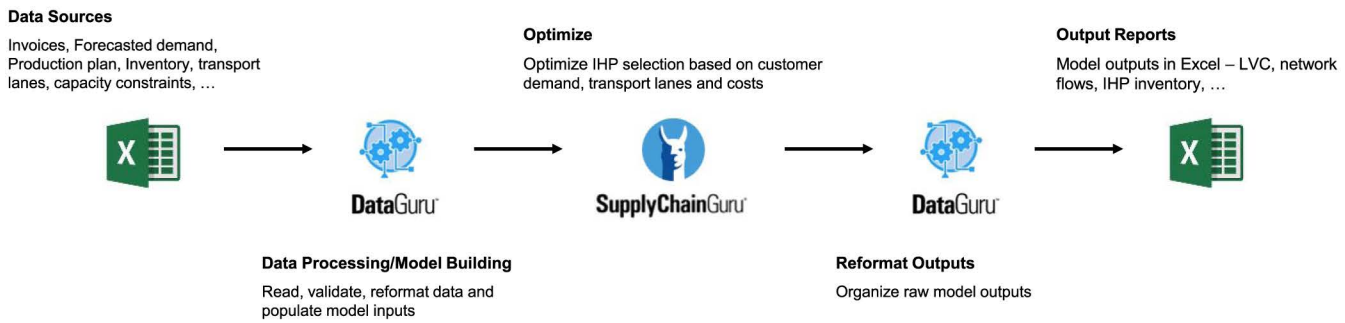

Figure 5—data ingestion and transformation workflow overview

The model will require several data sets to be gathered and populated as per master template files (Excel) required by SCGX. The various files required fall into three broad categories such as master data, transactional data and transport rates. Master files include data related to IHPs, customer and country mappings, ports, SKUs, packing and production capacities and storage constraints. Transactional data includes historical invoices, forecasted demand, costs per component of each IHP, stock transfers, on-hand inventory and production plans. The transport rates files include ocean container and road costs from all possible from and to location combinations within the model. Every month, these files are stored in a specific location on the VM, and the DG process is run to convert them into inputs that can be fed into the SCGX model.

The advantage of using Data Guru is that the output excel files are pushed into Supply Chain Guru interface without needing to open the model and manually importing the required excel files as they have a common backend. Hence, after data transformation in Data Guru, the next step in the process is to simply

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run the model followed by additional data transformations within DG again, to convert the raw output excel files created by SCGX into specifically formatted reports required for operations.

InPo Logic (to reduce model complexity). One of the key pieces of transformation to reduce the number of ports (InPos) from 390 to roughly 211, termed as the InPo logic has been implemented within Data Guru. The first step is to create a summary of all possible combinations of customer and it's corresponding InPo with the total quantity converted in homogeneous Unit of Measures that the InPo has supplied for a given SKU. This is then sorted in a descending order and a percentage of the total quantity supplied by an InPo to customer is calculated. Two conditions are applied as a test to each InPo and customer combination, for a given SKU. The first checks if the total quantity supplied by that InPo to the customer should be more than m% of the total quantity to that customer and the second checks If this InPo is among the top 20 InPos supplying to that customer. If both the conditions are satisfied, then that InPo is included else, it is eliminated. Note that the value of m varies for different customer groupings and can be seen in the table below in the 'Count' and 'InPo Cutoff' columns. The percentage threshold is used to group customers by their total cumulative demand into a specific category. As an example, China accounts for a total of 24% of global demand and so is within the threshold of 70% and so is a category green customer for which, any InPo supplying less than 0.4% of the total quantity to China is ignored. Note that the Count column lists the number of customers within each category, reaching a total of 112 customers.

% Threshold	Category	Count	InPo Cutoff
0%	Green	8	0.4%
70%	Yellow	20	2%
95%	Blue	16	10%
98%	Red	68	40%

Figure 6—InPo Cutoff lookup table for given customer categories.

This procedure, which reduces the total number of InPos from 390 to roughly 211, causes only 3% of the total volume to be ignored when modeling and helps towards reducing model complexity. A screenshot of the InPo logic as seen within the DG interface is shown below:

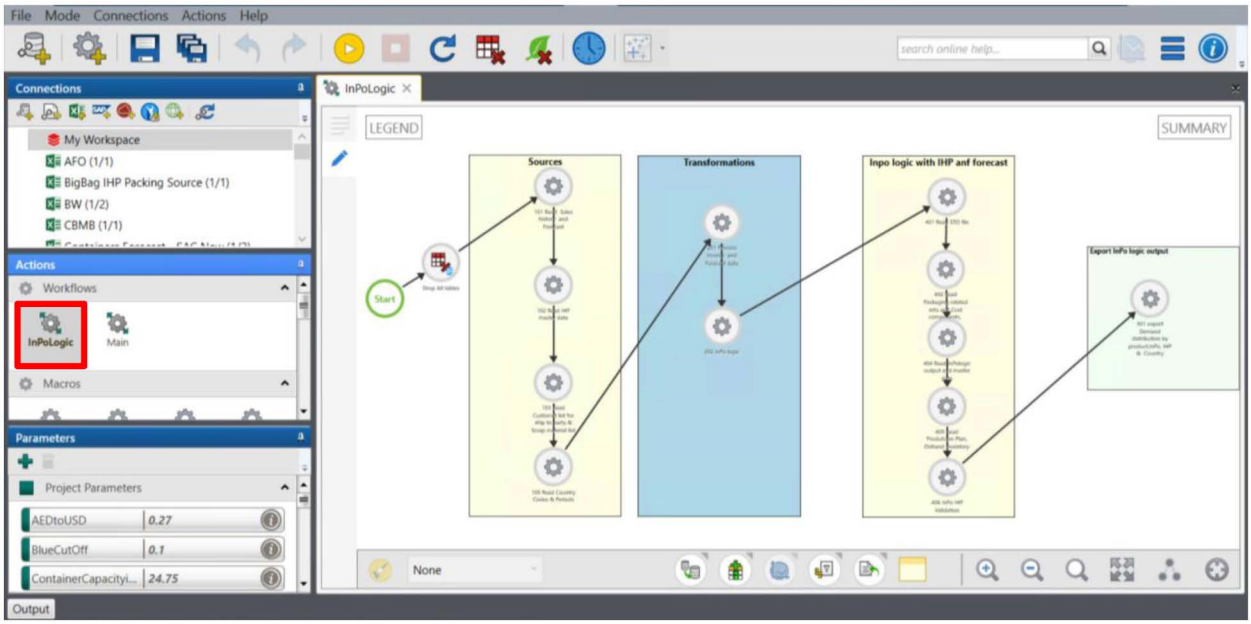

Figure 7—InPo Logic transformation steps within DG

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Creating the baseline model. The model was first used to develop the baseline which would reflect current plans for the next 12 months. This will allow the model to generate a base for comparison when running multiple scenarios. It is important to first develop an optimal network scenario which disables replicating product flows as per the baseline to provide a view of what could be achieved before implementing constraints in other scenarios. The optimal network scenario along with the different scenarios already developed will be explained in the scenarios section within this paper.

Model echelons. The baseline model consists of four echelons, i.e. the plant, IHPs, followed by the ports (InPo) and then the end customers. Historical movement of all products through all the lanes traversed in historical transport data, beginning from the plant and ending at the destination port have been duplicated to form a baseline model. There are four echelons of transport in the global supply chain network, and they are the plant to UAE IHPs (road, sea freight and rail), followed by UAE IHPs to international IHPs (sea freight), followed by all IHPs to destination ports (sea freight) and finally from the destination ports (InPo) to customer. However, note that costs in the last leg are not modeled in the network.

Model costs. The various costs that have been modeled are the packing costs (only available at the Al Ruwais plant and two other IHPs), handling costs, storage costs and transport costs. The former three costs together form part of fixed and variable costs associated with the sites in the models (plant, IHP and InPo) and the latter cost is associated with the transport legs within the model. Thus, all the costs used in transportation, production and associated costs with the IHP and InPos have been modelled and when added up form the total cost of the baseline scenario, Model constraints

Constraints have been applied to packing (production constraints table), transport lane capacity (flow and flow count constraints table), inventory capacity (inventory constraints table) and transit times (Transportation policies table). Also, the live inventory position is also loaded into the baseline and is the beginning inventory for PERIOD_00 which is the first period within the model (PERIOD_11 being the last).

Miscellaneous configurations. Note that a dummy plant beside the Al Ruwais facility, named as MFG_0000 has also been configured into the model so that demand which cannot be satisfied due to specific constraints is then satisfied by the dummy plant. The volumes that are satisfied by the dummy plant needs to be minimal, to conclude that the baseline is satisfactorily working.

Scenarios. The final stage in developing the model is creating scenarios which are broken down into two sections, Scenarios and Scenario Items. Scenarios consist of a collection of scenario items. A scenario item identifies the action that will be taken to adjust the model outcome, either constraining the model or relaxing it, to allow the model to act in a way that the modeler feels suits the needs of their operations. As part of the network optimization project, several scenario items and scenarios were developed.

Scenario example (Optimize IHP selection). Note that a scenario item consists of changes made to a single table within the model. Multiple scenario items are thus needed to implement a single scenario. As an example, consider the second scenario in the table below. It's purpose is to optimize the baseline scenario by removing restrictions on the network. Currently when creating the baseline, the modeler needs to make sure that all flows that are keyed in, need to mimic the actual flows that have happened so as to arrive at a total cost that is reasonably close to the actual cost incurred for the supply chain network. But these flows may not be intuitive when it comes to saving costs, and hence any constraints that have been applied (which are generally implemented on multiple tables within SCGX), need to be disabled. Coming back to our example, all flows which mimic the baseline are stored within the flow constraints table and hence a scenario item is created for this purpose. Next, there are flow count constraints which set limits on how many sites a customer can source from or how many plants a site can source from, etc. These can also be relaxed so that the model can arrive at a solution that is not limited to the flow counts imposed on it to mimic the baseline flows. To summarize, the 'Optimize IHP selection' scenario consists of two scenario items, one which disables baseline flows and the other disables baseline flow count constraints.

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Summary of scenarios. The scenarios implemented in the model are summarized below. The description column lists out the scenario items created within each scenario.

Table 1—Scenarios implemented within the model

Sr. No.	Scenario	Description	Comment
1	BaselineNoLT	All lead times from dummy plant to IHPs are set to zero	Used for debugging infeasibility
2	Optimize IHP selection	Baseline flows disabled & flow count constraints disabled	All baseline constraints removed, thus enabling optimization
3	Optimize IHP + CN constrained	Constraints of (2) and inventory constraints for CN warehouses	Optimized scenario with an inventory limit on China warehouses
4	Optimize IHP + CN constrained + M0 lock	Constraints of (3) and M0 locked wrt flow and flow count constraints	Optimized scenario with inventory limit on China & month zero to behave as per plan
5	Optimize IHP for specific customer	Similar to (4) except that constraints for a specific customer are disabled	It is optimized for a single customer with the rest like baseline
6	UAE vs overseas IHP split	Constraints of (4) and flow constraints setting a ratio of UAE IHPs vs overseas IHPs	Flow constraints modified so that demand is served in a specific ratio between UAE and other IHPs
7	Modify site throughput and inventory capacity	Constraints of (4) and a throughput capacity + inventory capacity limit on specific IHPs	Used to add throughput and inventory capacity to specific IHPs
8	Modify InPo ratio for selected customers	Constraints of (4) + exclude existing baseline customer sourcing policies + add specific customer sourcing policies	Override existing port to customer demand ratio served
9	IHP single source for customer at product-period level	Constraints of (4) + flow constraints to enforce single sourcing for a customer	To limit specific customers to use a single source
10	DIT vs WH utilization split at customer-product level	Constraints of (4) + flow constraints to enforce a split between DIT and WH to a customer	To enforce a specific percentage of demand of a customer to be served by DIT and WH IHPs. Note that an IHP is classified as a warehouse or a DIT (port storage).
11	Exclusion of selected transportation policies	Constraints of (4) + specific transport policies excluded	To exclude specific lanes
12	Scenario (4) + (7) + (10)	Constraints related to (4) & (7) and (10)	Combining three scenarios into one

Once these scenarios were created, the baseline model was run, along with these scenarios and the total cost for each scenario is displayed below.

Outputs and visualizations

Standard model outputs. The outputs created by SCGX are in the form of tables which can be exported as Excel files. The output tables broadly have two categories – summary and flow (transactional) tables. Summary tables take one aspect of the supply chain and summarize required transactions and attributes by that aspect. Examples are network summary (summary of total profit, costs and revenue), network customer summary, network site summary, network product summary, network inventory summary, customer demand summary and constraint summary. As an example, the network site summary summarizes all transactions (product flow, costs, open/close status, etc.) by all the sites within the network, whereas the customer demand summary summarizes all transactions (total demand, demand served, costs, etc.) and customer attributes by each individual customer within the network. Transactional tables contain each transaction at a row level and examples are customer flows, inter-site flows and production flows. As an example, customer flows contain data grouped at customer, site, period and product level, with each row signifying the product moved within a specific period from a specific site to a specific customer whereas inter-site flows are similar but contain product flows from site to site.

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Modified outputs using DG. However, these standard output tables may require some level of data transformations to convert them into standardized reports fit for internal use and hence, DG has been used again to convert these standard tables into specifically formatted Excel reports. Examples are Lost Demand Report, LVC report, Network Cost Report, Transport Cost Report RWS to IHP and Transport Cost Report IHP to InPo. The Lost Demand report simply details any demand that is not satisfied. The LVC report breaks down all costs with each flow at an origin, material, destination and period level. The network cost report summarizes costs by each scenario and finally the two transport cost reports detail volumes and costs for lanes between the Al Ruwais plant and the IHPs and IHPs to InPos respectively.

Output visualizations. Next, output visualizations have been created in a QlikView environment which includes standard visualizations reporting on various aspects of the supply chain network such as costs, volume, etc. as well as maps showing the various flows within the supply chain.

Challenges encountered and approach used. A critical element in supply chain design projects is the need to validate intermediate steps before proceeding to subsequent ones. An example of this is the need to validate and approve the data set that will be used in the model after the initial data has been aggregated, manipulated, harmonized, and completed. Another example is the need to validate the baseline before proceeding to scenario building. It often happens that those responsible for approving these intermediate steps have difficulty doing so if they do not have a perfect understanding of the nature of the project, the technology, and the assumptions that have been made.

To address this issue, the approach followed was to accompany the project's development with continuous training and the ongoing transfer of skills and knowledge. This approach allowed decision-makers to approve intermediate steps by providing them with all the necessary information and knowledge required for an informed and straightforward evaluation and validation of both intermediate and final steps.

Another critical point in the development of these projects is that very often the scope or assumptions defined initially need to be revised during the project due to the complex and dynamic nature of the supply chain design process. To adhere to the initially defined timelines and budget, a continuous, attentive, and flexible project management methodology was adopted. This methodology focuses on budget and time control while respecting the project's overall objectives. A robust, repeatable planning process was implemented along with knowledge transfer to enable continuous and constant monitoring of activity progress. This allows for adjustments when necessary, ensuring maximum awareness and speed from the team responsible for making these decisions.

Another important aspect to consider in these types of projects is the need to involve all levels of the organization from the outset to fully understand the process and development methodology. There is a risk in such projects that as activities progress, the value may not seem immediately apparent. In fact, value is identified only at the end of the project during the optimization phase related to the development of alternative scenarios. To reduce the risk associated with the intrinsic nature of this methodology, it is crucial to involve all stakeholders so that the initiative has the organization's support, which understands its nature, timelines, and methodology.

It is also important to inform all involved parties throughout the project phases about the nature of the technology used. This technology differs from classic reporting or cost calculation systems due to its ability to add intelligence to basic information to develop optimization scenarios that can support both strategic and tactical decisions. From this perspective, it is essential to understand the need to shift from the level of detail typical of transactional systems that analyze data at the maximum level of detail to a model that operates at a higher level of approximation using aggregation and simplification criteria.

Another important aspect to consider is that in some phases of the project, it is possible to parallelize certain activities. For example, a thorough evaluation of the baseline for its validation at the highest level of detail can be paralleled with the beginning of optimization scenario construction. This approach allows

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maintaining a project plan within the initially defined horizon while extending the expected duration of some specific activities.

Another challenge was understanding complex operational level challenges like production lines, packaging capacities and converting them to strategic level requirements in the network. Next, optimizing the ports to be considered in the network out of thousands of ports used from historical data to few hundreds to reduce the complexity was an important task. Also, we needed additional support from their planners to correct the product plan to match the future demand and remove correct the imbalances. There were multiple scenario requirements from different business users, and so the team had to come up with minimal building blocks required which can in combination help them to create most of their scenarios.

Conclusions

Implementing global supply chain network as a digital twin has unlocked multiple benefits such as having a cost associated with each echelon of the network, having access to multiple reports and views of the current as-is network as well as 'simulated' networks, having a method to quantitatively assess the advantages and disadvantages of choosing a different network strategy over an existing network, and an ability to detect in advance any issues that may arise in the course of the 12 month period of the network model besides other advantages.

Due to the nature of the implemented model, the stakeholder can drill down into the costs and other attributes of the model to provide the inner workings of the network in the form of reports and visualizations in a much shorter time than before. Without the digital model, the stakeholder would have to transform raw data from scratch into the specific report required. Multiple what-if scenarios can be run and compared alongside the baseline scenario in terms of costs, service level and other attributes if so required. This gives a quantitative method to evaluate and go ahead with a different strategy as opposed to the current one. Also, as the model is a multi-period model, the reports coming out of the model will give a 12-month future view of the flows and operations within the network and can be used to alert stakeholders should there be any bottlenecks that appear in periods in the future such as shortage of inventory at a specific note, unavailability of material to satisfy demand, etc.

As also mentioned within the results sub-section and the image below, the optimized baseline models (scenarios 5, 10, 11 and 12) show a cost savings of around 9%.

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